Last time Counting

MP, $P(n,k)$, $\binom{n}{k}$

formula/identity $P(n,k) = \binom{n}{k} k!$ "Counting" argument

arrange k of n objects; ① choose k of n objects; ② arrange k chosen objects

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Lesson 12 Combinatorial Proofs

Prop 1 If $|A| = n$ (A a set), then $|\mathcal{P}(A)| = 2^n$.

$\mathcal{P}(A)$: the set of all subsets of A

(If A has 3 elements, A has $2^3 = 8$ subsets)

Proof To count the # of subsets of $A = \{a_1, \ldots, a_n\}$ we think of a subset as a series of decisions to include or exclude each element a_k, $k = 1, \ldots, n$.

Aside

$A = \{1, 2, 3\}$

$\{1\} \longleftrightarrow (1, 0, 0)$ — 3-element binary list (0, 1)

1: include the 1st element; 0, 0: exclude the 2nd and 3rd elements.

$\{2\} \longleftrightarrow (0, 1, 0)$

$\{1, 3\} \longleftrightarrow (1, 0, 1)$

Each subset corresponds to a n-element binary list.

# of subsets = # of $\overline{\text{n-element binary lists}}$

$$= \underbrace{2 \times 2 \times \cdots \times 2}_{n} = 2^n$$

$$\underset{0,1}{\overset{2}{\_}}\ \overset{2}{\_}\ \_\ \cdots\ \_\ \_$$

Prop 2 Let n be a nonnegative integer, then

$$\underbrace{\sum_{k=0}^{n} \binom{n}{k}}= 2^n$$

identity/ formula

$$\left[\underset{k=0}{\binom{n}{0}} + \underset{k=1}{\binom{n}{1}} + \underset{k=2}{\binom{n}{2}} + \cdots + \underset{k=n}{\binom{n}{n}}\right]$$

algebraic proof : $\binom{n}{k} = \dfrac{n!}{(n-k)!k!}$

(combinatorial)

Proof From Prop 1, the RHS, 2^n, is the # of subsets of an n-element set A. We will that the sum on the LHS also counts the # of subsets of A.

Recall that $\binom{n}{k}$ is the # of k-element subsets of an n-element set.

A has $\binom{n}{0}$ subsets w/ 0 element

A .. $\binom{n}{1}$ ... w/ 1 element

$\cdots\cdots \binom{n}{2} \cdots\cdots$ 2 elements

$\vdots$

$\cdots\cdots \binom{n}{n} \cdots\cdots$ n elements

The total # of subsets of A is

$$\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\cdots+\binom{n}{n}=\sum_{k=0}^{n}\binom{n}{k}$$

Binomial Theorem $n \in \mathbb{Z}, n \geq 0$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

binomial coef

$n=0: (x+y)^0 = \binom{0}{0} x^{0-0} y^0 = 1 \cdot 1 \cdot 1 = 1$

$n=1: (x+y)^1 = \binom{1}{0} x^{1-0} y^0 + \binom{1}{1} x^{1-1} y^1$

$= x+y$

$n=2: (x+y)^2 = \underset{\binom{2}{0}}{x^2} + \underset{\binom{2}{1}}{2xy} + \underset{\binom{2}{2}}{y^2}$

$n=3: (x+y)^3 = \underset{\binom{3}{0}}{x^3} + \underset{\binom{3}{1}}{3x^2y} + \underset{\binom{3}{2}}{3xy^2} + \underset{\binom{3}{3}}{y^3}$

Pascal's Triangle

$$
\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}
\qquad
\begin{array}{ccccccccc}
 & & & & \binom{0}{0} & & & & \\
 & & & \binom{1}{0} & & \binom{1}{1} & & & \\
 & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & \\
 & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & \\
\binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4}
\end{array}
$$

proof next time

$$
\begin{cases}
\binom{n}{k} = \binom{n}{n-k} \\
\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}
\end{cases}
$$

Binomial Thm:

$((a)+b)((c)+d)$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

Proof $(x+y)^n = \underbrace{(x+y)(x+y)\cdots(x+y)}_{n}$

When expanded, we have a sum of terms $x^i y^j$ because we get exactly one x or y from each factor $(x+y)$. Moreover, $i+j=n$.

So, if there are k y's,
we have $(n-k)$ x's.
How many $x^{n-k} y^k$ terms?
The answer is the # of ways
we choose k y's from n factors,
that is, $\binom{n}{k}$.
Finally we sum from $k=0$ to n
because we can have $0, 1, 2, \ldots, n$ or n
y's in each term.